

Sienna Camper Conversion

A modular, tool-free lift-out sleeping + galley system for a Toyota Sienna — sleeps two, with a slide-out fridge and camp kitchen, EcoFlow power and A/C, and no permanent modification to the van. This page summarizes every feature and shows the drawings, so you can skim it and flag anything you'd change.

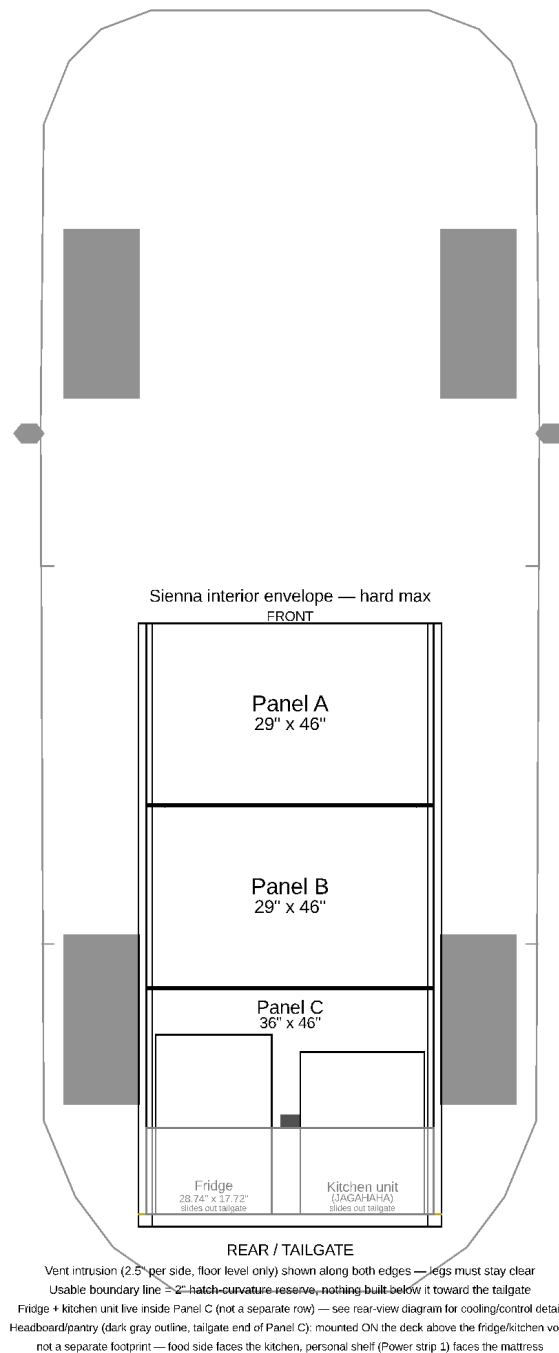
How to give feedback: skim the seven drawings and the feature list, then jump to [Open questions](#) — those are the calls I'd most like a second opinion on (and the measurements still unverified). Tell me what's missing, what you'd do differently, or where the design looks fragile.

SLEEPS 2 52"-wide platform, 25"/person	VAN INTERIOR 96 × 48.5 × 44" L × W × H (length est.)	MODULES 3 + 1 Panels A/B/C + clamp-on headboard	POWER DELTA 3 + Extra Battery · WAVE 3 A/C
COLD 41 QT Rocky 40 dual-zone, slide-out	EST. COST \$3.1–3.5k full; ~\$2.4k if gear owned		

THE LAYOUT

One continuous deck of three lift-out boxes; the fridge, kitchen, and pantry all share Panel C at the tailgate end.

BIRD'S-EYE FLOOR PLAN

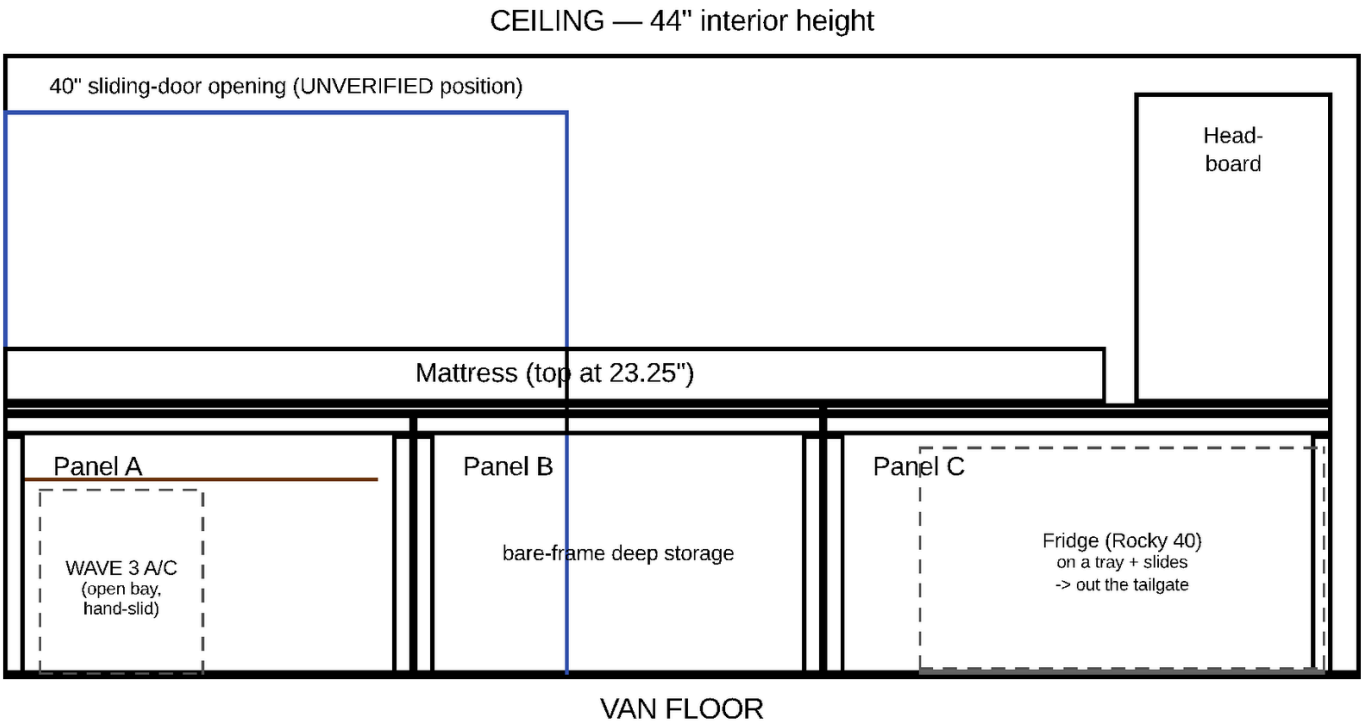


Top-down. Front of the van at top, tailgate at bottom. Panel A (29") + Panel B (29") carry the bed; Panel C (36") holds the fridge (driver side) and kitchen (passenger side) underneath, with the pantry/headboard mounted on the deck above them. Grey blocks = wheel wells; the platform stays clear of the floor-level heat vents at the edges.

THE DRAWINGS

The two side elevations show what each sliding door reaches; the two gate views show the galley stowed for driving vs. deployed for use.

DRIVER SIDE

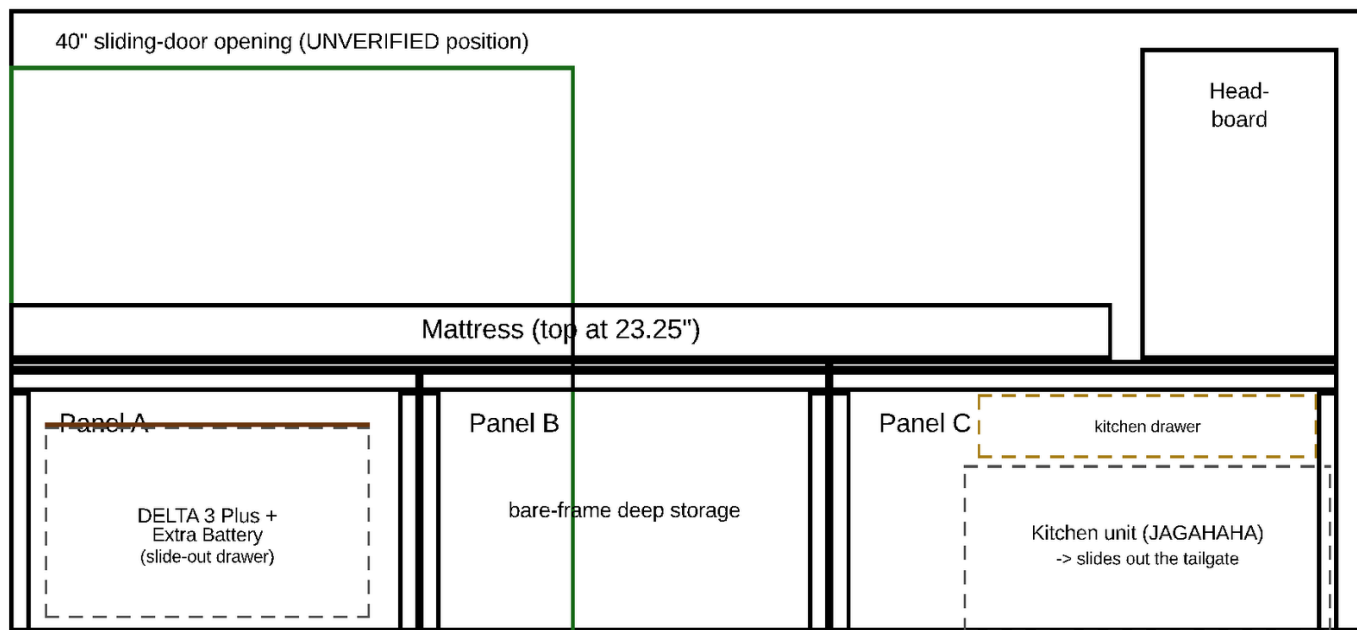


DRIVER SIDE — looking in from this door. FRONT SEATS at left, TAILGATE at right.
This side: WAVE 3 (Panel A) + fridge (Panel C). Dashed = under-deck items reached from here.

From the driver door. WAVE 3 A/C in Panel A's open bay; the fridge under Panel C. Dashed = under-deck items reached from this side.

PASSENGER SIDE

CEILING — 44" interior height



VAN FLOOR

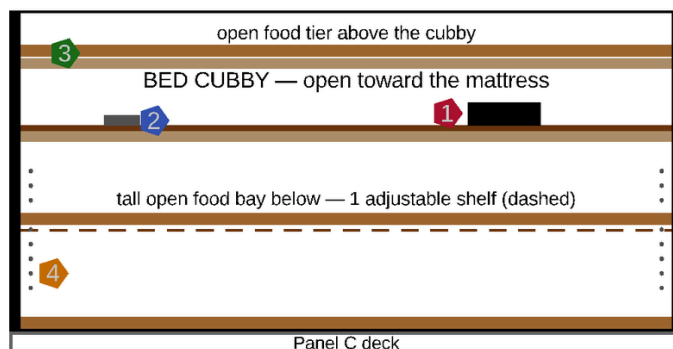
PASSENGER SIDE — looking in from this door. FRONT SEATS at left, TAILGATE at right.
This side: DELTA 3 power (Panel A) + kitchen unit (Panel C). Dashed = under-deck items reached from here.

From the passenger door. The DELTA 3 power drawer in Panel A; the kitchen unit + its hung drawer under Panel C.

LOOKING BACK TOWARD THE HEADBOARD

LOOKING FROM THE FRONT SEATS BACK TOWARD THE HEADBOARD (bed side)

46" wide x 22" tall, on Panel C's deck — NO top panel; food side faces the tailgate



Legend

- 1 Power strip 1 (bed-cubby)
- 2 ROLL bubble level (leveling)
- 3 Fiddle lip on every food tier
- 4 Shelf-pin columns (adjustable bay)

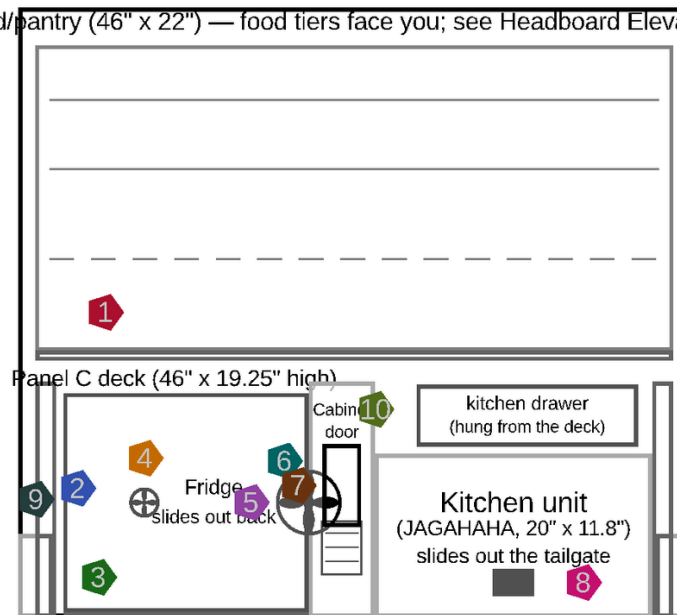
Bed-cubby floor sits 9" above the mattress top, 2.75" deep, full width — open only toward the mattress.
Enclosed by a food tier above + a tall food bay below; the nook divider forms its ceiling. Every food tier: fiddle lip + lash strap.

From the bed. The pantry's mattress-facing side: an enclosed bed cubby in the middle band (phone shelf + Power strip 1 + a bubble level, 9" above the mattress), an open food tier above it and a tall adjustable food bay below. Food side faces the tailgate; no top panel.

INTO THE GATE — STOWED

Sienna interior envelope (width x height) — hard max

Headboard/pantry (46" x 22") — food tiers face you; see Headboard Elevations



Legend

- 1 Frame / deck / fridge tray
- 2 Kitchen unit (standalone)
- 3 Fridge (standalone)
- 4 Intake fan (120mm)
- 5 Exhaust fan (120mm)
- 6 NTC temp sensor
- 7 Control panel
- 8 Power strip 2 (cooktop)
- 9 Floor vent intrusion (both sides)
- 10 Utility cabinet door

Power strip 2 (cooktop)

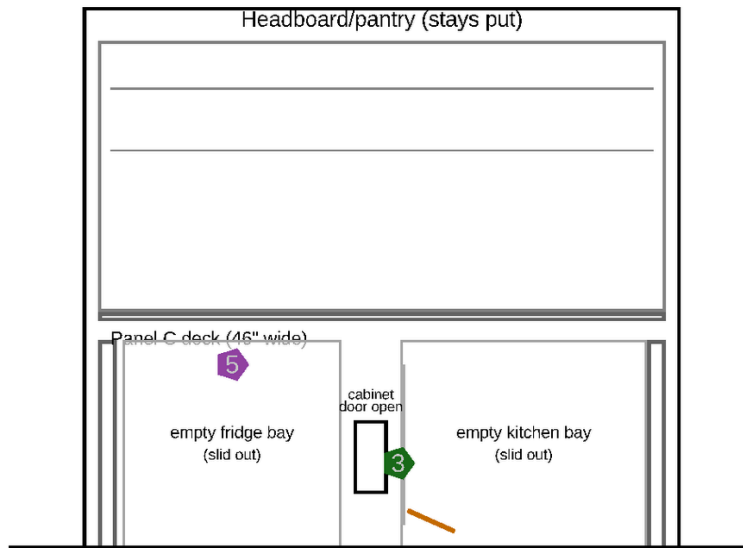
Exhaust fan: fridge's right wall, blows INTO the cabinet | NTC probe: just inside the bay at that wall (in the hot exhaust, NOT the cabinet)
Intake fan + a passive LOW cool-air louver: both on Panel C's FRONT wall (see its render) | cabinet door has a LOW exhaust louver
Control panel: switches + surge protector — INSIDE the cabinet, behind the door

Looking forward from the open tailgate at Panel C — both units shown stowed for driving

Driving mode. Fridge + kitchen tucked under Panel C's deck; cooling fans, NTC sensor and control panel live in the utility cabinet between them.

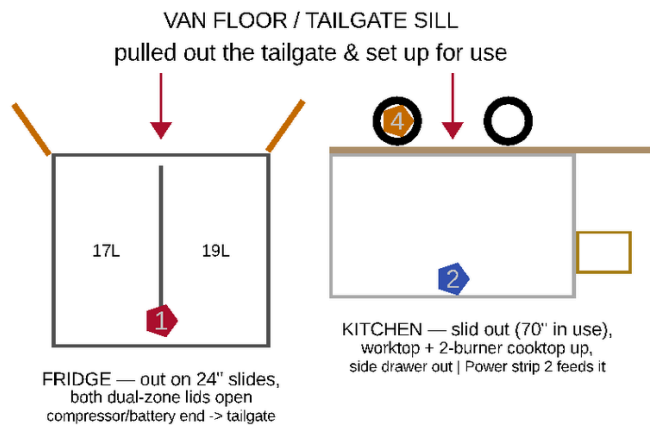
INTO THE GATE – DEPLOYED

Sienna interior envelope (width x height)



Legend

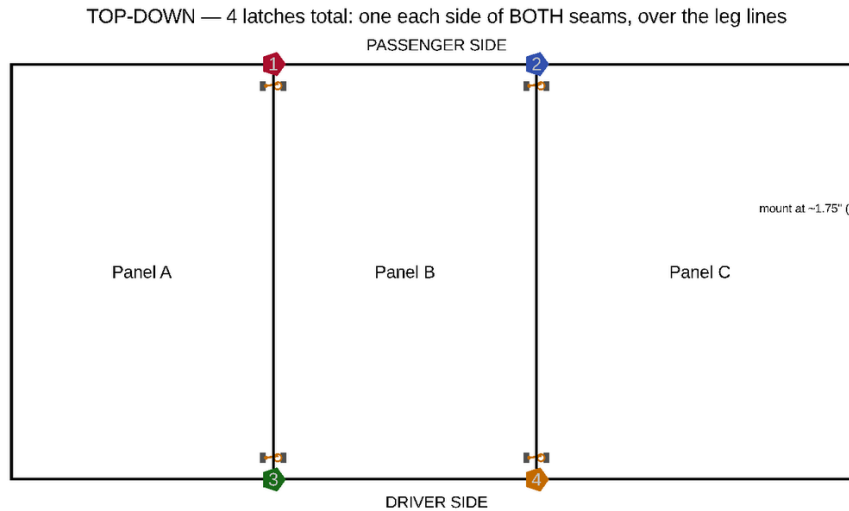
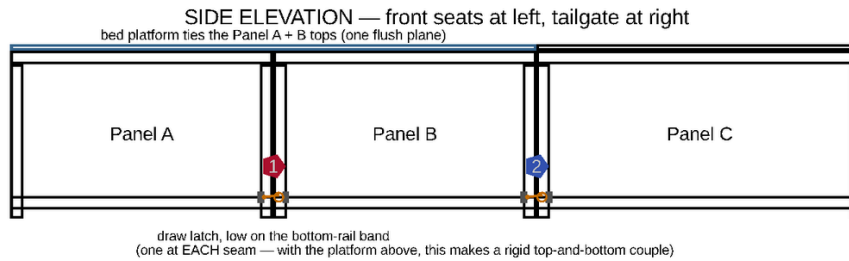
- 1 Fridge — deployed, both lids open
- 2 Kitchen unit — deployed, cooktop up, drawer out
- 3 Utility cabinet (door open): control panel + fans
- 4 2-burner induction cooktop on the worktop
- 5 Empty bays left behind in Panel C



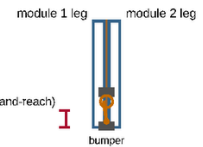
DEPLOYED — looking into the open tailgate with the fridge & kitchen pulled out and set up for use

Camp mode. Both slide out the tailgate: fridge lids open (dual-zone 17L+19L), kitchen worktop + 2-burner cooktop up and drawer out. The bays behind them sit empty.

SEAM DRAW-LATCH POSITIONING



DETAIL — over-center draw latch
base plate on one leg, hooked bail + lever on the other;
flip the handle to clamp tight, flip to release (no tools)



SEAM DRAW-LATCHES — clamp the 3 lift-out modules into one rigid beam (4 latches: both sides of the A/B and B/C seams)
Alignment pins LOCATE the modules; the draw latches PULL them tight against the bumper. Hand-released — modules still lift out in seconds.

Clamping the modules. The alignment pins only *locate* the three boxes; four hand-released over-center draw-latches — one on each side of both seams, low on the bottom-rail band — pull them tight into one rigid beam. With the bed platform tying the tops, the low latches complete a top-and-bottom couple that kills sway and rattle, while any panel still lifts out in seconds.

FEATURES

What each part is and why it's there. Amber notes flag the ones I'd most like feedback on.

STRUCTURE **Modular lift-out platform**

Three independent boxes — Panel A (29"), B (29"), C (36"), all 46" wide — built on 2×2 pine frames with leveling legs, plus a clamp-on pantry. Nothing bolts to the van; each box lifts out through the liftgate as a one/two-person carry. Cube-frame bottom rails add racking stiffness, and **4 hand-released over-center draw-latches** at the seams clamp the three boxes into one rigid beam without giving up the lift-out.

Built weight is $\approx$ **424 lb** (after lighter-wood swaps), ~884 lb loaded with two people — comfortably under the van's payload, though the mass is rear-biased (see the weight budget in the plan).

SLEEPING **Cantilevered bed platform**

A 52"-wide slatted 1×4 platform spans Panels A+B and ends flush at the B/C seam, meeting Panel C's own deck. It overhangs the 46" boxes by 3" per side — using the fact that the van is wider up at mattress height than at the pinched floor.

Feedback wanted — this depends on $\geq 53"$ of wall-to-wall width at 19–23" above the floor, which isn't measured yet (see question 3).

SLEEPING **Mattress**

HEST Dually Long — 78 × 50 × 4" solid foam with a washable waterproof cover, 25" per person. Leaves roughly 16.75" of sitting headroom to the ceiling.

GALLEY **Headboard / pantry**

A 46 × 22" shelving wall clamped onto Panel C's deck with four cam levers — tool-free, flips off in a minute. Food tiers (fixed + one adjustable shelf) face the kitchen; an enclosed cubby faces the mattress. Retention is a marine-galley system — a fiddle lip on every shelf plus lash straps, bins and non-slip liner — rather than plain bungee nets.

Feedback wanted — it's a 22" wall right at the tailgate opening; confirm it clears the liftgate's swing and doesn't block reaching the appliances as they slide out (question 6).

GALLEY Fridge

BougeRV Rocky 40 — dual-zone 41 QT, 40.6 lb, 60W max — on a heavy-duty 24" slide + plywood tray, pulled out the open tailgate for lid access. The compressor/battery end faces the tailgate so the optional battery swaps without a full slide-out.

GALLEY Kitchen unit

JAGAHAHA slide-out camp kitchen (2-burner, own drawer), strapped to floor anchors, sliding out the tailgate and extending to ~70" in use. A shallow kitchen drawer hangs in the dead air above it for utensils and flat goods. The induction cooktop runs off Power strip 2.

COOLING Forced-air fridge bay

Two 120mm fans — intake on Panel C's front wall, exhaust on the fridge's kitchen-facing wall blowing into the utility cabinet — on a W1209 thermostat reading an NTC probe in the hot exhaust, so they only spin when the compressor is working. Two passive louvers (low intake + cabinet exhaust) back them up. This compensates for the bay being tighter than the fridge's passive-venting spec.

POWER EcoFlow system

DELTA 3 Plus + Smart Extra Battery ride in Panel A's slide-out drawer (charged at camp via shore power/solar, not from the van). The WAVE 3 air conditioner stores in Panel A's open bay and runs at the tailgate or the front seat. Two AC power strips feed the cooktop and the bed cubby.

LEVELING Two-stage leveling

Level the van at the wheels with Lynx blocks, guided by a phone Block Calculator that reads two bed-mounted bubble levels and tells you blocks-per-tire. The interior leg feet are then set once against the van's own floor. (Electric feet were considered and rejected.)

STORAGE**Found space + joinery**

Reclaimed dead headroom — a tray over the DELTA 3 stack, a shelf over the WAVE 3, bins in the cabinet gap — and Panel B is a bare-frame deep-storage cube. The bottom stabilization frame was dropped lower for more usable volume. Carcasses use Ryobi DBJ50 biscuit joints (R-series) for strength and clean faces, with fastener locations dimensioned.

SAFETY**Air & alarms**

An owner-placed low-mounted CO monitor (audible from the bed) and a fire extinguisher, plus SRS seat-delete emulators so the airbag system reads normally with the 2nd row removed.

OPEN QUESTIONS FOR REVIEWERS

The unverified measurements and judgment calls. Items tagged **VERIFY** need a tape measure on the actual van before anything is cut.

1 Interior length **VERIFY**

96" is an estimate for the empty interior with the 2nd row removed. Measure closed-hatch to front seatbacks — the whole assembly is sized to fill it exactly.

2 Liftgate opening **VERIFY**

48 × 36" is web-sourced, not measured. Every module must pass through it upright, and the upper corners round off — measure the real narrowest width and height.

3 Bed cantilever width **VERIFY**

The 52" platform overhangs the 46" boxes and needs ≥53" wall-to-wall at 19–23" above the floor along the whole run. If it's tighter, the platform shrinks.

4 Side-door position & reach **VERIFY**

The door's fore-aft position is a placeholder. It decides which drawers are actually reachable from the side vs. blocked by body structure.

5 **Rear-corner floor vents** VERIFY

Panel C's rear legs sit at the true corners; confirm the floor heat vents don't reach the last ~4" there.

6 **Headboard vs. liftgate swing** VERIFY

The 22" pantry wall sits at the tailgate. Confirm it clears the gate's swing-up and that the fridge/kitchen still reach out past it.

7 **Appliance specs & price** VERIFY

Confirm live-listing dimensions and current price for the Rocky 40 fridge and the JAGAHABA kitchen before ordering.

8 **Floor anchors**

The E-track anchors bolt through the floor pan — inspect underneath for fuel/brake lines and wiring, and seal each hole.

DESIGN CALLS WORTH A SECOND OPINION

Beyond the measurements, these are open on the merits — I'd value a view:

- Is a **slide-out kitchen + separate fridge** the right split, or would one integrated galley be simpler and stronger?
- Is **two 120mm fans + passive louvers** enough fridge cooling for real summer heat, or should the bay get a bigger fan or a dedicated vent?
- Clamp-on pantry at the tailgate vs. moving it forward — does having it right at the gate get in the way in practice?
- Anything **missing entirely** — water/sink, interior lighting, a shore-power inlet, insulation, screens?

COMPANION TOOLS

The interactive/printable sheets that go with this plan.

Block Calculator →

Blocks-per-tire from the bubble levels

Shopping List →

Materials, quantities & links

Cut List →

Every board & panel, grouped

Build Sequence →

Dependency-ordered assembly

Drawings are to-scale OpenSCAD line art from the parametric model; every dimension is computed from the plan's source of truth, not sketched. Full detail lives in **Project_Smores.pdf** (Components 1–10 + appendices). Estimates and unverified figures are flagged throughout.